Avery Karlin
Columbia University
New York, NY, USA

June 21, 2026

Professor John R. Griffiths
Editor-in-Chief, *NMR in Biomedicine*

Dear Professor Griffiths,

I am pleased to submit "Boundary-railing of conventional NLLS fits as an assumption-free pseudo-diffusion identifiability diagnostic in IVIM MRI" for consideration in *NMR in Biomedicine*.

I want to be transparent about the history of this work, because the correction is part of what makes it worth publishing. An earlier version led with a calibration claim — a dramatic marginal under-coverage of the pseudo-diffusion coefficient D^* — and a methods review rightly objected on two grounds: that an empirical coverage statement is only as trustworthy as the noise and forward model behind its reference truth, and that the dramatic figure depended on choices that were not documented. We took both objections seriously and rebuilt the contribution.

Two things changed. First, an independent clean-room reimplementation showed that the dramatic *marginal* headline does not reproduce; it is recoverable only under an undocumented, overconfident convention for the standard deviation of a railed (unidentified) voxel. We have dropped that headline. The honest, reproducible statement is *conditional*: under a stated Cramér–Rao convention, symmetric Gaussian intervals under-cover D^* specifically in the high-D^* tercile (central-95% coverage 0.63) while remaining near-nominal elsewhere; a shape-aware quantile interval restores the marginal coverage but not the conditional residual, and an amortized neural posterior dominates the railed baseline. The convention that manufactured the original severity is now documented in full, alongside all dataset identifiers, fitting and training detail, and the Cramér–Rao approximation.

Second — and this is the heart of the resubmission — we promote to the primary claim a result that *cannot* be overextended because it assumes nothing. A conventional box-constrained least-squares IVIM fit drives D^* onto a parameter bound in 54.7% of high-SNR voxels on the open OSIPI abdomen scan (95% CI 52.2–57.1), reproduced by the independent reimplementation (54.2%) and replicated across the full ROI (47.8%) and an independent liver DWI cohort (43.7–73.4%). This boundary-railing is read directly off the optimiser, with no ground truth and no noise-model trust, and so it answers the reviewer's central objection rather than defending against it.

The result is a stronger, narrower, and fully reproducible paper: a primary claim immune to the overextension critique, and a calibration result kept only within the scope its evidence supports. Every number traces to a single seeded reproduction that regenerates with one command on openly licensed data (OSIPI TF2.4, Zenodo 14605039; TCGA-LIHC via The Cancer Imaging Archive). The work fits both principal categories of *NMR in Biomedicine* and is complementary to learned-uncertainty frameworks for IVIM such as Casali et al. (2026, this journal), who themselves report residual D^* overconfidence — the learned echo of the same identifiability wall.

The manuscript is original, has not been published previously, and is not under consideration elsewhere. The author declares no competing interests, takes full responsibility for the integrity of the work, and includes a declaration of AI-assisted writing. I would be glad to suggest reviewers and to answer any questions.

Sincerely,

Avery Karlin, BA
Department of Applied Physics and Applied Mathematics, Columbia University
ak5232@columbia.edu